Physical Adversarial Attacks Against Aerial Object Detection With Feature-Aligned Expandable Textures

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***Abstract*— Physical adversarial attacks in aerial object detection have gained significant attention. Existing adversarial patches exhibit subpar visual effects and encounter limitations when transitioning from digital to physical spaces, restricting applicability in real-world scenarios. To address these challenges, we propose an adversarial texture generation method based on background texture design. This method selectively covers the background environment without interfering with the target surface. We also explore the translational invariance of fully convolutional networks to decouple adversarial textures from shapes, allowing adversarial textures to be arbitrarily expanded during use. The areas where adversarial textures are placed are designated as the "detection failure zone," rendering the detector ineffective regardless of the aircraft's position within this zone. This significantly enhances the practicality of the adversarial texture. To improve its concealment, we align the features of the adversarial textures with those of the original image using a pretrained VGG network, ensuring a consistent style and color tone with the background environment. Additionally, we employ a discriminator to further control the visual effects of the adversarial samples, ensuring effective concealment. Furthermore, we simulate the real environment in digital space using operations like affine transformations and Gaussian blur to transfer adversarial textures seamlessly from digital to physical space. This allows for the integration of adversarial textures into real environments without compromising their effectiveness. Experimental results demonstrate the effectiveness and consistent styling of the proposed adversarial texture in real-world environments, showing robustness against environmental changes, weather conditions, and viewing angles.**

***Index Terms*— Adversarial textures, aerial detection, convolutional network, feature alignment, remote sensing.**

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I. Introduction

OBJECT detection stands as a pivotal technology extensively utilized in safety–critical scenarios, including industrial control and aerospace, especially in remote sensing [1]. Integrating deep neural networks (DNNs) into object detection has resulted in a substantial enhancement of detection accuracy [2], [3], [4], [5], [6]. Nevertheless, the intrinsic vulnerability of DNNs introduces novel challenges to the dependability and security of object detection methodologies grounded in this technology [7], [8], [9], [10], [11], [12]. Research on novel adversarial attack algorithms establishes a foundational framework for enhancing the adversarial robustness of DNNs and offers insights into their susceptibility to adversarial samples [10], [13], [14], [15], [16].

Current physical adversarial attacks on remote sensing aerial detection typically focus on disrupting target objects, such as painting or attaching patches to aircraft or ships [17]. However, these attacks depend on specific conditions like lighting, angles, and distances, leading to variable effectiveness and requiring meticulous design. For nonplanar targets, applying textures to a 2-D image plane creates significant disparities between the optimized and actual processes [18], [19], [20]. Large-scale application is costly and time-consuming, and foreground adversarial textures are often conspicuous, raising suspicion. Background textures also play a crucial role in aerial detection [21], [22]. If the background texture resembles the target or is intricate, detection algorithms may struggle to distinguish between the target and background, leading to missed detections. Existing adversarial patches based on background textures are often visually prominent and heavily reliant on texture structure and color [23]. Replicating these textures in the physical world is challenging, and issues like occlusion and fading reduce robustness in dynamic environments [22], [24].

Therefore, we propose an adversarial attack algorithm that utilizes expandable, visually harmonious, and robust background textures for object detection in airport remote sensing scenarios, as shown in Fig. 1. In terms of practicality, unlike traditional adversarial patches [22], [24], our method generates adversarial textures with relatively simple structures that can be freely cut and spliced. Theoretically, these textures can be "infinitely extended" to cover an entire airport without disrupting its structural integrity. As long as the aircraft hovers

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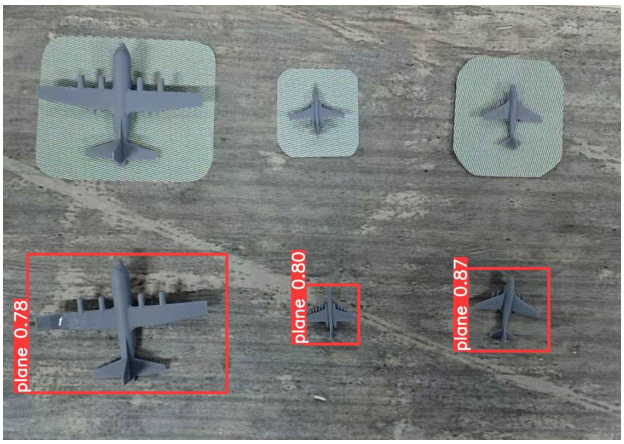


Fig. 1. Schematic of physical adversarial attacks includes a "detection failure zone" where adversarial textures are placed. The upper section shows YOLOv5s detection results for an aircraft in this zone, while the lower section displays results for a clean sample.

or lands on the adversarially textured ground, it can effectively attack the target detector from any position. Visually, the generated adversarial textures match the color tone of the airport environment, with softer colors. The brightness and contrast of the textures are consistent with the original images, enhancing their concealment. Regarding robustness, our adversarial samples demonstrate higher robustness through real environment simulations. They maintain strong attack effectiveness under various weather conditions and scenarios involving wear and fading.

Technically, we propose a novel physical attack framework based on background information, aiming to decouple the generated adversarial textures' geometric regularity, i.e., separating the adversarial nature of the texture from its shape. This enables arbitrary stitching or cropping of adversarial textures without compromising their adversarial effectiveness. To harmonize the chromatic information of the adversarial texture with the airport background, we align the feature information extracted from the original images with that of the adversarial texture during the sampling process, ensuring distribution consistency [25]. The generated adversarial texture maintains stylistic consistency with the original image, seamlessly blending into the airport environment and enhancing concealment. To boost the robustness of the adversarial texture in real-world settings, we comprehensively simulate the physical environment in the digital space using techniques like affine transformations, Gaussian blurring, and random noise addition. This holistic approach aims to minimize the gap between digital and physical spaces. The integrated methodology enables the generated adversarial textures to adapt to diverse airport environments and exhibit a certain level of attack effectiveness against different aircraft models. Our method has been rigorously tested with leading object detection frameworks. Additionally, we conduct tests in real airport sandbox environments to confirm the practical viability of our proposed method. Experiments in both digital and physical domains affirm the robustness of the adversarial textures produced by our method.

The contributions of this article are summarized as follows:

1) We propose a framework for generating specialized physical adversarial textures for backgrounds. Leveraging the translational invariance of convolutional networks, our resulting adversarial texture is isotropic, allowing for seamless concatenation and cutting of locally optimized textures. We align features at different levels with the background distribution, ensuring expandability and consistency with the airport style.
2) Within this framework, we build a simulator for real environments to comprehensively simulate physical reality, enhancing the feasibility of adversarial textures. We simulate aspects of the real physical environment in the digital space, incorporating color domain mismatches and diverse weather conditions. Techniques like affine transformations, random cropping, Gaussian blurring, and more are applied to replicate these conditions, augmenting the physical transferability of the adversarial texture.
3) Our adversarial textures exhibit stronger robustness. We explored the impact of adversarial attacks in diverse conditions, considering varying sizes and degrees of fading in the texture. The results highlight strong attack efficacy, especially when the adversarial texture's area is smaller than that of the aircraft or when a certain degree of fading is present. Additionally, our method demonstrates effective attacks across diverse airport environments and various aircraft models.

The rest of the article is organized as follows. Section II introduces the related works of adversarial examples. In Section III, details of the proposed method are introduced. Section IV provides experiments on the effectiveness and rationality of the method. Finally, we conclude the method in Section V.

II. Related Work

In this section, we explore the core concepts of adversarial attacks and review classical algorithms in image recognition. Subsequently, we focus on research into adversarial attack methods for object detection, particularly emphasizing algorithms for remote sensing scene object detection, and briefly introduce style transfer, a component in constructing the adversarial texture generator.

A. Adversarial Attacks

Early research on adversarial samples focused on data-level manipulation in image recognition tasks, categorizing attacks into white-box and black-box methods. White-box attacks leverage access to the victim model's structure and parameters [26], [27], employing gradient-based techniques (e.g., FGSM [8] and PGD [15]) and data training (e.g., AdvGAN [28]). Black-box attacks operate with limited or no internal model access, employing algorithms like transfer-based attack methods, gradient-based attack methods, decision-based attack methods, GAN-based attack methods, and other black-box attack methods [29]. Adversarial attacks

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are further categorized into target and nontarget attacks. Research on adversarial attacks has expanded to object detection, borrowing from image classification advancements [18], [30], [31], [32]. Target attacks aim to manipulate the victim model into outputting a predetermined class. However, in object detection, which includes both classification and additional outputs (e.g., bounding box or instance segmentation mask), targeting becomes challenging in high-dimensional spaces, particularly when the desired category significantly differs from the original. Nontarget attacks aim to make the model output any class other than the original, typically constructing adversarial samples toward the nearest decision boundary [33]. In summary, the procedure of adversarial attacks grounded in image recognition can be delineated as follows [34]:

$$\min_{x'} D\left(x, x'\right) \quad \text{s.t. } f\left(x'\right) \neq y(x) \tag{1}$$

where $y(x)$ represents the true label of the original image x and x' represents the adversarial sample. $f(\cdot)$ is the victim model, and $D(x, x')$ is an evaluation metric for the visual dissimilarity between the adversarial sample and the original image. Most digital-domain adversarial attacks emphasize the imperceptibility of perturbations by introducing subtle disturbances globally or locally to influence model predictions [35], [36], [37]. However, applying such perturbations directly to the physical world is generally challenging. Consequently, physical attack methods have been gaining attention [38], [39].

B. Physical Attacks

Physical adversarial attacks involve reproducing adversarial textures generated in digital images in the physical world through printing devices or spray paint. Kurakin et al. [39] were the pioneers in discussing adversarial samples in the physical world, uncovering the presence of adversarial samples with noticeable attack effects.

For face recognition, Sharif et al. [40] conducted attacks by printing a pair of eyeglass frames. Wei et al. [41]introduced natural adversarial stickers, designing physically present stickers to deceive facial detection systems when applied to a face. Additionally, adversarial hats, masks, and even manipulated lighting conditions have been devised to mislead face recognition systems [17], [19], [42], [43].

For human detection, Thys et al. [44] introduced the Adversarial-YOLO method in 2019, specifically designed to generate adversarial patches targeting the YOLO model. The patches generated by this method can compromise the performance of the YOLO-based pedestrian detection system, making it unable to detect pedestrians. Xu et al. [20] presented a physical adversarial example using an adversarial T-shirt to evade object detection models. Considering variations in human poses and different scenes, they proposed a thin plate spline (TPS) method to simulate T-shirt deformations, demonstrating effective attack performance in both digital and physical domains. Huang et al. [45] introduced the Universal Physical Camouflage (UPC) method, creating universal adversarial patches capable of inducing targeted outputs from object detection models for specified categories and achieving targeted attacks. Doan et al. [46] introduced the Tnt method, generating a universally natural adversarial patch. This method employs generative adversarial networks to learn semantic information from real images in the natural world, generating a patch with specific patterns and colors, and achieving universal targeted attacks. Hu et al. [24] propose the Toroidal-Cropping-based Expandable Generative Attack (TC-EGA) method to create AdvTextures with repetitive structures for items like T-shirts and skirts, allowing attacks on human detectors even with changing viewing angles. However, despite the repetitive structures, the textures have poor visual quality and the method does not address the impact of occlusion or varying weather conditions on attack performance.

For automatic driving systems, Lu et al. [47] crafted adversarial samples for "stop signs," successfully deceiving a physical-domain traffic sign detection model, albeit with suboptimal visuals. Zhang et al. [48] introduced CAMOU (Camouflages), concealing vehicles through deceptive patterns. Utilizing generative adversarial networks, CAMOU learns the mapping from target images (e.g., vehicle photos) to camouflage patterns, challenging computer vision systems in vehicle detection. The adversarial texture exhibits high transferability, proving effective across diverse environments, vehicles, and detection models. Zolf et al. [49] proposed the Translucent Patch method, disrupting object edges and textures with real-world translucent films, rendering object detection systems unable to identify objects accurately.

C. Physical Adversarial Attacks for Aerial Image Object Detection

DNNs are widely applied in processing aerial images and understanding adversarial samples aids in explaining and boosting model robustness [50], [51]. However, most attack algorithms against aerial image detectors primarily focus on the digital domain, with limited research on physical attacks [27], [52], [53], [54]. Nevertheless, physical attacks are more practical. Du et al. [55] were the first to achieve physical attacks on aerial images by placing adversarial patches on target objects or near them, thereby reducing the effectiveness of detectors applied to aerial image object detection. Lian et al. [33] introduced a novel Physical Patch Attack framework (APPA), generating adversarial patches adaptable to physical dynamics and varying scales. They also designed a new loss function that optimizes the patch based on the detector's detection results. Building upon foreground-targeted physical attack methods, Lian et al. [7] proposed a Contextual Background Attack (CBA) framework on top of APPA to enhance the adversarial texture's attacking effect. This framework masks foreground targets with a designed mask, optimizes pixels outside the masked region, and ensures that the generated adversarial patch tightly covers critical contextual background areas of the detected targets. The CBA framework achieves strong attack capabilities and transferability without contaminating foreground targets. However, the visually suboptimal adversarial patches generated by this method make it challenging to extend to real airport environments. Wang et al. [22] propose dual-adversarial-induced

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error identification (BADEI), a framework that deceives detectors by swapping backgrounds and targets. This enhances attack performance through error identification in both scenarios. However, it also encounters the issue of visual harmony with adversarial samples.

III. Proposed Method

In developing the adversarial texture generator, we face two primary challenges. First, ensuring that the adversarial texture is free from constraints related to texture shapes and structures, displaying expandability, and maintaining alignment with the airport background style. Second, addressing the need to preserve the properties of the texture during its transition from the digital to the physical domain and ensures robustness during application. Our algorithm flowchart is shown in Fig. 2.

A. Problem Formulation

In aerial detection tasks, given a clean aerial image x, the purpose of adversarial texture is to induce omissions or false positives in the target detector for the adversarial sample x'. The process of constructing adversarial samples can be represented as follows [33]:

$$x' = \left(1 - M_{p'}\right) \odot x + M_{p'} \odot p' \tag{2}$$

where $\odot$ and p' respectively denote the Hadamard product and the adversarial texture. The mask matrix $M_{p'}$ is employed to constrain the size, shape, and position of the adversarial texture, ensuring its adherence to the airport background and avoiding interference with foreground targets. Further details on acquiring background adversarial textures with robust adversarial effects and a certain degree of visual harmony are discussed in Section III-B.

B. Construction of Adversarial Texture Generator

Our objective is to generate expandable, visually coherent adversarial textures. The region covered by the texture within the airport is referred to as the "detection undefined zone," where placing the aircraft at any position within this zone can influence the detector's results. These adversarial textures exhibit robustness in real-world environments and a high degree of similarity to the airport style.

Exploiting the translation invariance property inherent in convolutional networks, we utilize a fully convolutional network, denoted as K, for the construction of the adversarial texture generator [24], [56], [57]. This network is predominantly composed of zero-padded convolutional layers, with the initial layer dedicated to handling input latent variables. Inspired by Adaptive Instance Normalization (AdaIN) [25], [58], [59] to enhance the concealment and stylistic consistency of the adversarial texture with the airport style, we adjust the mean and variance of the texture feature maps based on those of the original image. A feature alignment layer, embedded within the fully convolutional network, aligns the adversarial texture with the channel mean and variance of the original image. The multichannel feature map of the adversarial texture, denoted as $\boldsymbol{P}$, is expressed as follows [58], [60]:

$$\boldsymbol{P} = \left[\boldsymbol{p_1}, \boldsymbol{p}_2, \ldots, \boldsymbol{p_C}\right] \in \mathbb{R}^{H \times W \times C} \tag{3}$$

where $\boldsymbol{p_n} \in \mathbb{R}^{H \times W \times 1}$ refers to the nth column vector of $\boldsymbol{P}$, representing the vectorized feature map of the nth channel with dimensions $H \times W$. Additionally, the feature map $\boldsymbol{X}$ of the original image is denoted as follows:

$$\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_C] \in \mathbb{R}^{H \times W \times C}. \tag{4}$$

Subsequently, we utilize a pretrained network to align the features extracted from the original image with the corresponding channels during the upsampling process of adversarial textures. In other words, we adjust the mean and variance of the adversarial texture according to those of the original image

$$\Gamma(\boldsymbol{P}, \boldsymbol{X}) := \sigma(\boldsymbol{x})\left(\frac{\boldsymbol{p} - \mu(\boldsymbol{p})}{\sigma(\boldsymbol{p})}\right) + \mu(\boldsymbol{x}) \tag{5}$$

where $\mu(\boldsymbol{p})$ and $\sigma(\boldsymbol{p})$ represent the mean and variance, respectively, calculated independently for each channel and each sample in the spatial dimension for the generated texture. $\mu(\boldsymbol{x})$ and $\sigma(\boldsymbol{x})$ signify the mean and variance of the features extracted from the original image [61]. This process involves normalizing the texture features to remove their original statistical properties, followed by rescaling and shifting to match the statistical properties of the background features [25], [58], [59]. By integrating the feature alignment layer into the texture generator, unity is achieved in the feature space between the adversarial texture and the original image style. The final expression for the adversarial texture generator can be formulated as follows:

$$p' = K\left(z, \theta_g, x, \Gamma(\boldsymbol{P}, \boldsymbol{X})\right) \tag{6}$$

where the latent variable $z \sim \mathcal{N}(0, I)$ is a tensor of size $\mathbb{R}^{B \times C \times H \times W}$, where B stands for the batch size, C represents the number of channels, and H and W are the height and width, respectively. θ_g denotes the parameters of the adversarial texture generator and x signifies the original image.

C. Facilitating Physical Realizability

During adversarial texture generation, a mismatch exists between digital and physical spaces, involving factors like resolution, camera noise, and lighting conditions [20], [62], [63]. This mismatch renders adversarial samples generated in the digital space ineffective in attacking detectors in the physical space [64]. Hence, in the adversarial texture generation process, we introduce random variations to simulate lighting changes at different times, weather conditions, or scenes. Concurrently, we simulate physical effects such as sensor noise and lens distortion by adding random noise. Additionally, we apply random affine transformations to the images to adapt to different angles and directions of cameras in the real world. To accommodate varying camera sensor resolutions, we downsample and apply Gaussian blur to the images. These processes comprehensively simulate the physical environment, enabling adversarial textures generated digitally to effectively transition into the real world. By incorporating and simulating physical environmental factors, this approach enhances the adaptability and attack effectiveness of adversarial textures in real-world scenarios.

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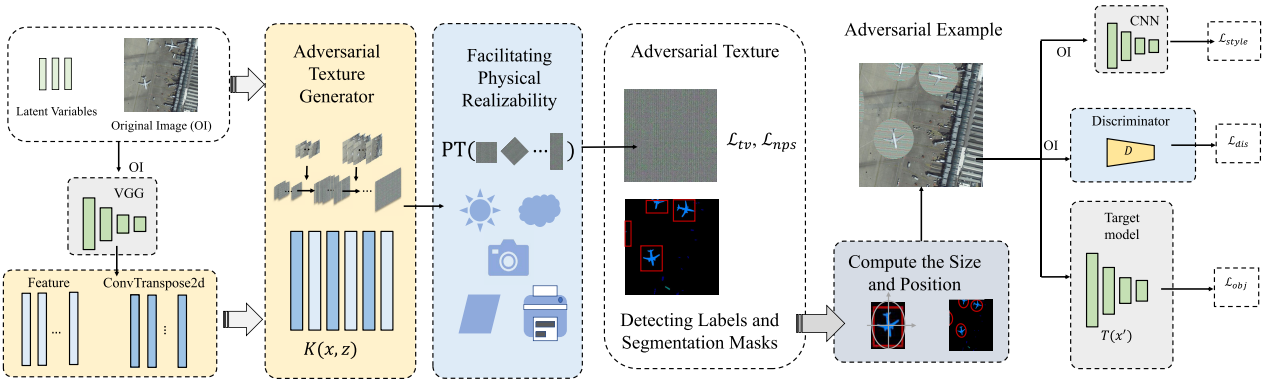


Fig. 2. We introduce an adaptive method for adversarial texture generation through feature alignment. Initially, an adversarial texture generator is constructed with convolutional networks and feature alignment layers, utilizing randomly generated latent variables and original images as inputs. Pretrained VGG networks are employed for feature extraction from the original images. During upsampling, feature alignment ensures the alignment of the distribution of adversarial texture features with those of the original images. Subsequently, the generated adversarial textures undergo simulation in a realistic environment. Enhanced diversity and printability are achieved through the application of L_{tv} and L_{nps}. Following this, the placement and size of adversarial textures are determined based on instance segmentation masks and object detection labels. To mitigate the impact of texture shape boundaries on adversarial effects, elliptical cropping is applied, utilizing the aircraft's long and short axes. Finally, adversarial samples are fed into the object detector, where detection confidence is the loss function L_{obj}. Furthermore, to enhance the visual quality of adversarial samples, we introduce L_{style}, and integrate a discriminator, imposing additional constraints, L_{dis}, in the optimization process.

D. Patch Adapting

We encapsulate the physical adaptive operations in Section III-C within the adversarial texture transformation function PT(·). The adversarial sample can be further rewritten as follows

$$x' = \left(1 - M_{p'}\right) \odot x + M_{p'} \odot \text{PT}\left(p'\right). \tag{7}$$

Next, we elliptically crop the generated texture to liberate the adversarial texture from direction constraints. We then concentrate on computing the shape and position of the adversarial texture, determining the center coordinates $C_(p')$ based on the bounding box result's coordinates (d_x, d_y, w, h). Here, d_x and d_y denote the normalized coordinates of the bounding box center concerning image width and height, while w and h refer to the width and height of the bounding box

$$C_{p'} = \left(c_x, c_y\right) = \left(d_x, d_y\right). \tag{8}$$

Then, taking into account the differing proportions of wings and fuselages among various aircraft, the selection of the semimajor and semiminor axes for the ellipse is as follows:

$$\begin{aligned} l_{p'} &= \frac{1}{2} r_{p'} \max(w, h) \\ s_{p'} &= \frac{1}{2} r_{p'} \min(w, h) \end{aligned} \tag{9}$$

where $r_{p'}$ serves as a hyperparameter regulating the size of the adversarial texture. Consequently, the standard equation for elliptical constraints on the shape and size of the adversarial texture is

$$F : \frac{\left(p'_x - c_x\right)^2}{{l_{p'}}^2} + \frac{\left(p'_y - c_y\right)^2}{{s_{p'}}^2} = 1 \tag{10}$$

where p'_x and p'_y denote the horizontal and vertical coordinates of the adversarial texture, respectively. Next, the mask $M_{p'}$ for the adversarial texture can be expressed as follows:

$$M_{p'} = F\left(p', C_{p'}, l_{p'}, s_{p'}\right) \tag{11}$$

To position the adversarial texture beneath the aircraft, we utilize the segmented mask $M_{p'}$ to extract the aircraft from the original image. Subsequently, we place the aircraft on the adversarial texture and then position the adversarial texture at the corresponding location in the original image. The original problem can thus be reformulated as follows:

$$\begin{aligned} x' = &\left(1 - F\left(p', C_{p'}, l_{p'}, s_{p'}\right)\right) \odot x \\ &+ r_o \cdot F\left(p', C_{p'}, l_{p'}, s_{p'}\right) \odot \text{PT}\left(p'\right) \odot M_p \odot x \end{aligned} \tag{12}$$

where r_o represents the adversarial texture overlay ratio, used to simulate phenomena such as paint fading and texture aging in the physical world.

E. Loss Function

In this article, our objective is nontargeted physical adversarial attacks, aiming to make the aircraft undetectable within the "detection undefined zone." To constrain the adversarial texture generator, the loss function primarily encompasses the following.

1) Det Attack Loss: The adversarial target loss is denoted as the average of all target scores detected after nonmaximum suppression [7]

$$L_{\text{obj}} = \frac{1}{n} \sum_{i=1}^{n} T_i(\text{obj}). \tag{13}$$

The target detection results, denoted by r, encompass the coordinates (x, y, w, h) of each object, the target score $T(\text{obj})$, and category scores for each class [e.g., $T(\text{airplane})$]. L_{obj} is the average confidence scores of n detected classes. The purpose of the adversarial texture is to conceal the target in the airport environment; hence, directly reducing the predicted target scores is the simplest and most effective approach [33].

2) Total Variation Loss: The total variation loss, denoted as L_{tv}, is incorporated into the loss function to address challenges arising from the difficulty of image acquisition devices in

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TABLE I
COMPARISON OF DIGITAL DOMAIN ATTACKS

Method	Model	Yolov5s	Yolov5n	Yolov5l	Faster R-CNN	Cascade R-CNN	Mask R-CNN	Swin-Transfomer*
CBA [7]	Yolov5s	**68.6%**	63.1%	74.3%	34.1%	34.1%	24.5%	66.7%
	Yolov5n	65.2%	**50.8%**	67.2%	31.1%	31.1%	21.1%	62.2%
	Yolov5l	71.5%	65.1%	**73.3%**	38.9%	28.9%	38.9%	66.8%
	Faster R-CNN	73.6%	66.1%	75.8%	**28.8%**	14.8%	29.8%	64.8%
	Cascade R-CNN	75.6%	67.1%	79.8%	28.8%	**18.8%**	28.8%	64.8%
	Mask R-CNN	74.6%	67.3%	78.8%	28.8%	14.8%	**28.8%**	64.8%
	Swin-Transfomer*	74.6%	72.3%	76.2%	30.0%	14.8%	29.6%	**60.4%**
Random Nosie [65]	Yolov5s	**58.60%**	-	-	-	-	-	-
	Yolov5n	-	**47.19%**	-	-	-	-	-
	Yolov5l	-	-	**71.86%**	-	-	-	-
	Faster R-CNN	-	-	-	**47.19%**	-	-	-
	Cascade R-CNN	-	-	-	-	**21.16%**	-	-
	Mask R-CNN	-	-	-	-	-	**57.48%**	-
	Swin-Transfomer*	-	-	-	-	-		**81.42%**
Random Texture [65]	Yolov5s	**85.25%**	-	-	-	-	-	-
	Yolov5n	-	**75.05%**	-	-	-	-	-
	Yolov5l	-	-	**93.02%**	-	-	-	-
	Faster R-CNN	-	-	-	**79.40%**	-	-	-
	Cascade R-CNN	-	-	-	-	**67.28%**	-	-
	Mask R-CNN	-	-	-	-	-	**86.32%**	-
	Swin-Transfomer*	-	-	-	-	-	-	**45.80%**
Pixel Noise [65]	Yolov5s	**58.08%**	46.57%	72.17%	14.64%	14.14%	13.68%	15.97%
	Yolov5n	8.64%	**4.29%**	9.98%	3.00%	3.57%	13.47%	4.25%
	Yolov5l	7.33%	4.96%	**11.44%**	13.97%	14.17%	13.57%	19.03%
	Faster R-CNN	14.65%	3.00%	13.97%	**1.66%**	2.38%	6.79%	3.93%
	Cascade R-CNN	14.14%	3.57%	14.17%	2.06%	**0.21%**	6.89%	0.62%
	Mask R-CNN	12.11%	4.79%	11.64%	6.79%	6.89%	**5.94%**	9.86%
	Swin-Transfomer*	15.97%	4.25%	19.03%	20.74%	20.02%	9.60%	**6.40%**
THREE-LAYER (Ours)	Yolov5s	**1.45%**	5.38%	38.36%	15.32%	3.32%	30.33%	17.85%
	Yolov5n	47.81%	**4.58%**	69.64%	20.50%	3.42%	30.75%	17.07%
	Yolov5l	39.47%	9.15%	**11.39%**	39.42%	24.45%	66.81%	61.91%
	Faster R-CNN	18.21%	7.93%	65.69%	**1.34%**	0.01%	4.14%	19.92%
	cascade R-CNN	28.18%	25.26%	65.69%	6.42%	**0.01%**	9.12%	17.44%
	Mask R-CNN	29.70%	26.20%	67.21%	1.55%	0.01%	**1.80%**	27.55%
	Swin-Transfomer*	37.95%	18.40%	67.00%	35.40%	11.56%	49.73%	**1.55%**
FIVE-LAYER (Ours)	Yolov5s	**1.71%**	5.38%	38.36%	42.75%	32.09%	46.45%	18.27%
	Yolov5n	41.71%	**1.87%**	60.52%	19.67%	4.44%	28.00%	6.81%
	Yolov5l	16.00%	9.19%	**7.55%**	42.86%	25.28%	63.73%	62.44%
	Faster R-CNN	18.21%	7.93%	65.69%	**1.35%**	0.01%	4.14%	19.92%
	Cascade R-CNN	29.54%	25.26%	65.60%	9.30%	**0.01%**	9.44%	17.92%
	Mask R-CNN	28.28%	27.55%	52.83%	1.45%	0.01%	**1.80%**	27.45%
	Swin-Transfomer*	36.62%	37.42%	61.63%	28.98%	5.88%	45.40%	**1.13%**

* The version used by the swin transformer is "mask-rcnn-swin-t-p4-w7-fpn-1x-coco.py".

capturing gaps between adjacent pixel values [40]. Additionally, disparities between real-world shooting angles/distances or deformations and the generation process of adversarial samples in the digital space may lead to attack failure. The introduction of the total variation loss aims to mitigate these issues, encouraging the optimizer to favor the generation of adversarial textures with smooth patterns and color shifts. The calculation of this loss function is as follows:

$$L_{\text{tv}} = \sum_{i,j} \sqrt{\left(v_{i+1,j} - v_{i,j}\right)^2 + \left(v_{i,j+1} - v_{i,j}\right)^2} \tag{14}$$

where $v_{i,j}$ represents the pixel value at the ith row and jth column of the adversarial texture.

3) Nonprintable Score Loss: Due to the limitations of physical printing and painting technologies, which may not achieve the same color gamut as digital processors, there is often a risk of information loss and color discrepancies. To address this, we impose constraints on the color gamut of the adversarial texture, introducing the printing loss L_{nps} [40]. This loss function quantifies the distance between the digital and physical domains, simulating a real-world printer. The specific representation is as follows:

$$L_{\text{nps}} = \sum_{i,j} \min_{C_{\text{print}} \in \text{Cor}} |C_{i,j} - \text{Cor}_{\text{print}}| \tag{15}$$

where $\text{Cor}_{\text{print}}$ represents one of the physically printable colors in a set of colors Cor, thereby constraining the pixel values $C_{i,j}$ of the adversarial texture to the collection of physically printable colors.

4) Style Loss: In enhancing the adherence of the adversarial texture to the airport style, we incorporate a style loss function L_{style} during the adversarial texture optimization process [66]. Employing a pretrained convolutional neural network, VGG19 [67], we represent the original image through feature maps at various levels. Each feature map is then flattened into a 1-D vector, and the subsequent computation involves the outer product of these vectors, resulting in the

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Gram matrix

$$G_{ij} = \sum_{k=1}^{H\times W} F_{ik} \cdot F_{jk} \tag{16}$$

where G_{ij} represents the correlation between feature maps F_i and F_j, and H and W denote the height and width of the feature maps.

We select five layers of feature maps, calculate their respective Gram matrices, and then compute the mean squared error among them. This mean squared error serves as the style loss function

$$L_{\text{style}} = \frac{1}{4C^2(H\times W)^2} \sum_{i,j} \left(G_{ij}^{\text{AE}} - G_{ij}^{\text{OR}}\right)^2 \tag{17}$$

where C is the number of channels, G_{ij}^{AE} represents the Gram matrix of the adversarial example, and G_{ij}^{OR} represents the Gram matrix of the original image.

5) Discriminative Loss: To further control the visual impact of the adversarial texture, we introduce a discriminative loss, denoted as L_{dis}, during the optimization of the adversarial texture generator. The discriminator D takes the adversarial sample as input and aims to discern whether the image is affected by adversarial perturbations [68]. Any adversarial sample exhibiting significant deviations from the original image or a substantially disrupted image structure will be identified by the discriminator, providing appropriate feedback to improve the visual quality of the adversarial texture. The applied discriminative loss is based on the binary cross-entropy (BCE) loss function and follows a min–max objective

$$\mathcal{L}_{\text{dis}}(D) = \mathbb{E}_x\left[\log D(x)\right] + \mathbb{E}_{x'}\left[\log\left(1 - D\left(x'\right)\right)\right]. \tag{18}$$

The adversarial texture generator and the discriminative network are trained jointly, progressively improving during optimization. The optimization objective can be formulated as the following min–max problem:

$$\begin{aligned}
&\min_K \max_D V(D,K) \\
&= \mathbb{E}_{x\sim r_{\text{data}}(x)}[\log D(x)] \\
&\quad + \mathbb{E}_{z\sim r_z(z)}\Big[\log\Big(1 - D\Big(\Big(1 - F\left(p', C_{p'}, l_{p'}, s_{p'}\right)\Big) \odot x \\
&\qquad + r_o \cdot F\left(p', C_{p'}, l_{p'}, s_{p'}\right) \\
&\qquad \odot \text{PT}\left(K\left(z, \theta_g, x, \Gamma(\boldsymbol{P}, \boldsymbol{X})\right)\right) \odot M_p \odot x\Big)\Big]
\end{aligned} \tag{19}$$

where K is the adversarial texture generator, D is the discriminative network, x represents the original samples, r represents the data distribution, and z denotes the latent variables for generating adversarial textures. This objective aims to minimize the distinguishability of generated adversarial textures from original samples while maximizing the discriminative capability of the network.

6) Total Loss: Since all components of our networks are differentiable, the whole framework can be trained in an end-to-end fashion. The total loss function is written as follows:

$$L_{\text{total}} = L_{\text{obj}} + \alpha \cdot L_{\text{tv}} + \beta \cdot L_{\text{nps}} + \gamma \cdot L_{\text{style}} + \delta \cdot L_{\text{dis}} \tag{20}$$

where α, β, γ, and δ are the hyperparameters for balancing different loss terms.

IV. Experiment

A. Datasets and Experimental Setup

1) Datasets: During the adversarial attack process, the simultaneous utilization of aircraft detection bounding box coordinates and aircraft mask information is essential. Thus, we have chosen to validate our approach using both the DOTA [69] and the iSAID [70].

DOTA [69] is a comprehensive open dataset designed for aerial object detection. It includes high-resolution satellite and aerial images from diverse regions, covering urban and rural areas under various weather and time conditions. The dataset spans multiple target categories, such as buildings, vehicles, aircraft, and vessels, capturing intricate details in different scenes. With rich annotations, including bounding boxes, category information, and occlusion details, DOTA serves as a benchmark for training and evaluating models in aerial object detection.

iSAID [70] improves upon the DOTA by introducing pixel-level annotations, establishing itself as the first benchmark dataset specifically designed for instance segmentation in aerial imagery. By rectifying annotation errors found in DOTA, there is a substantial increase in both sample size and annotation granularity. The meticulous alignment of target masks with visible edges in iSAID guarantees the provision of high-quality data for tasks related to instance segmentation. Here, we only utilize the segmentation masks from the dataset.

In our study, we focus on the airport environment in nadir-view aerial images, using a total of 157 images. During training, we experiment by segmenting images into 800 × 800 pixel dimensions with a coverage rate of 0.4. This setup aims to execute adversarial attacks, specifically targeting the aircraft category in the airport scene.

2) Target Models: We selected several representative object detection models, including one-stage detectors such as the YOLO series (YOLOv5s, YOLOv5n, and YOLOv5l) [4], two-stage detectors (Faster R-CNN [71] and Mask R-CNN [72]), Cascade R-CNN [73], and Transformer-based detectors [74], as the target victims and proxy models for the attacks. We extensively validated the effectiveness of the algorithm in both white-box and black-box settings.

YOLO [4] series of object detection algorithms achieve real-time detection in a single forward pass, emphasizing efficiency by integrating detection, localization, and classification into a unified network. Utilizing anchor boxes, they predict target positions and categories, continuously refining for improved accuracy and adaptability. The YOLO series has gained significant attention in computer vision for its real-time performance and efficiency.

R-CNN-Based Object Detection [75] is a region-based deep learning model designed for object detection. Its defining feature is the initial generation of candidate regions, followed by classification and bounding box regression. Notably, Faster R-CNN [71] improves upon the R-CNN series by introducing a Region Proposal Network (RPN) to replace traditional selective search methods in generating candidate regions. Mask R-CNN [72] extends Faster R-CNN with an additional branch for mask generation, enabling precise instance segmentation

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TABLE II
UNDER A VERTICAL PERSPECTIVE, EACH AIRCRAFT DETECTS CONFIDENCE USING YOLOV5S IN VARIOUS SCENARIOS

		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Average
Scene 1	Clean	0.92	0.92	0.83	0.38	0.88	0.93	0.88	0.93	0.89	0.80	**0.84**
	Texture 1	0.36	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.04**
	Texture 2	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Scene 2	Clean	0.78	0.91	0.92	0.88	0.76	0.93	0.94	0.91	0.81	0.87	**0.87**
	Texture 1	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 2	0.00	0.47	0.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.08**
	Texture 3	0.00	0.45	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.10**
	Texture 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Scene 3	Clean	0.72	0.82	0.79	0.76	0.85	0.79	0.89	0.85	0.87	0.81	**0.82**
	Texture 1	0.01	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.07**
	Texture 2	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 3	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Scene 4	Clean	0.91	0.88	0.87	0.87	0.91	0.91	0.89	0.87	0.92	0.91	**0.89**
	Texture 1	0.28	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.07**
	Texture 2	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**

alongside object detection. Similarly, Cascade R-CNN [73], derived from Faster R-CNN, introduces a cascading structure to progressively enhance object detection accuracy, particularly for small objects.

Transformer-Based Object Detector [74] applies the Transformer architecture to object detection tasks, replacing traditional convolutional neural networks. It achieves global context modeling of input sequences through self-attention mechanisms, allowing it to handle different scales and complex relationships more effectively in object detection. Additionally, it possesses strong parallel computing capabilities.

3) Metrics: The average precision (AP) is adopted as the quantitative detection metric. The detection results of clean images are used as the ground truth to calculate the AP, where the AP for the clean dataset is set to 100%. This ensures that the failure of the detector to detect targets in clean images is not considered a successful attack. The final calculation of the attack rate (AR) [7] is as follows:

$$\mathrm{AR} = \mathrm{AP}_{\mathrm{EX}}/\mathrm{AP}_{\mathrm{OR}} \tag{21}$$

where EX represents the adversarial sample, and OR represents the clean image.

B. Implementation Details

During the training phase, we adopt a one-stage training approach, involving random sampling of latent variables and optimizing the generator parameters. We configured hyperparameters α, β, γ, and δ as 2.5, 0.1, 1, and 0.5, respectively. The learning rates for the adversarial texture generator and discriminator are set to 0.001. The intersection-over-union (IOU) and the intersection threshold for objective confidence are set at 0.25 and 0.4, respectively. The model is trained using the PyTorch framework on RTX3090 (24 GB) GPUs. The sandbox simulates real environments at a 1:50 scale, including diverse terrains like urban areas, suburbs, mountains, and plains, and all adversarial textures are physically validated by printing them on copperplate article.

C. Analysis of Experimental Results

We conducted experiments in digital and physical domains to validate adversarial texture effectiveness. Diverse environments were used to assess texture robustness. For Sections IV-C1–IV-C3, a five-layer feature alignment architecture was employed for the adversarial texture generator. In Section IV-C4, we will analyze the distinct impact of feature alignment layers on the generated adversarial textures.

1) Attack Results: We validated our algorithm by comparing the state-of-the-art method CBA [7], random noise, random texture generation, and pixel-level optimization. In experiments, we varied adversarial texture generator architectures to confirm our approach's effectiveness. Different style alignment layers were used to assess attack effectiveness and visual coherence. Table I clearly shows that our method outperforms techniques using random noise, random textures, and pixel-wise optimization, whether using three-layer or five-layer feature alignment. Notably, adversarial textures generated by the YOLO detector effectively attack the R-CNN detector. However, adversarial textures from YOLOv5s and YOLOv5n exhibit lower effectiveness on the YOLOv5l detector, suggesting that transferability depends on both model structure and complexity.

Furthermore, insights from the table reveal nonlinear relationships in the attack effectiveness of adversarial samples on the original model and other models. In some cases, the original model displays strong attack effectiveness, yet the transferability of adversarial samples is weak. Insights from the table reveal nonlinear attack effectiveness on the original model and others. While the original model can be highly

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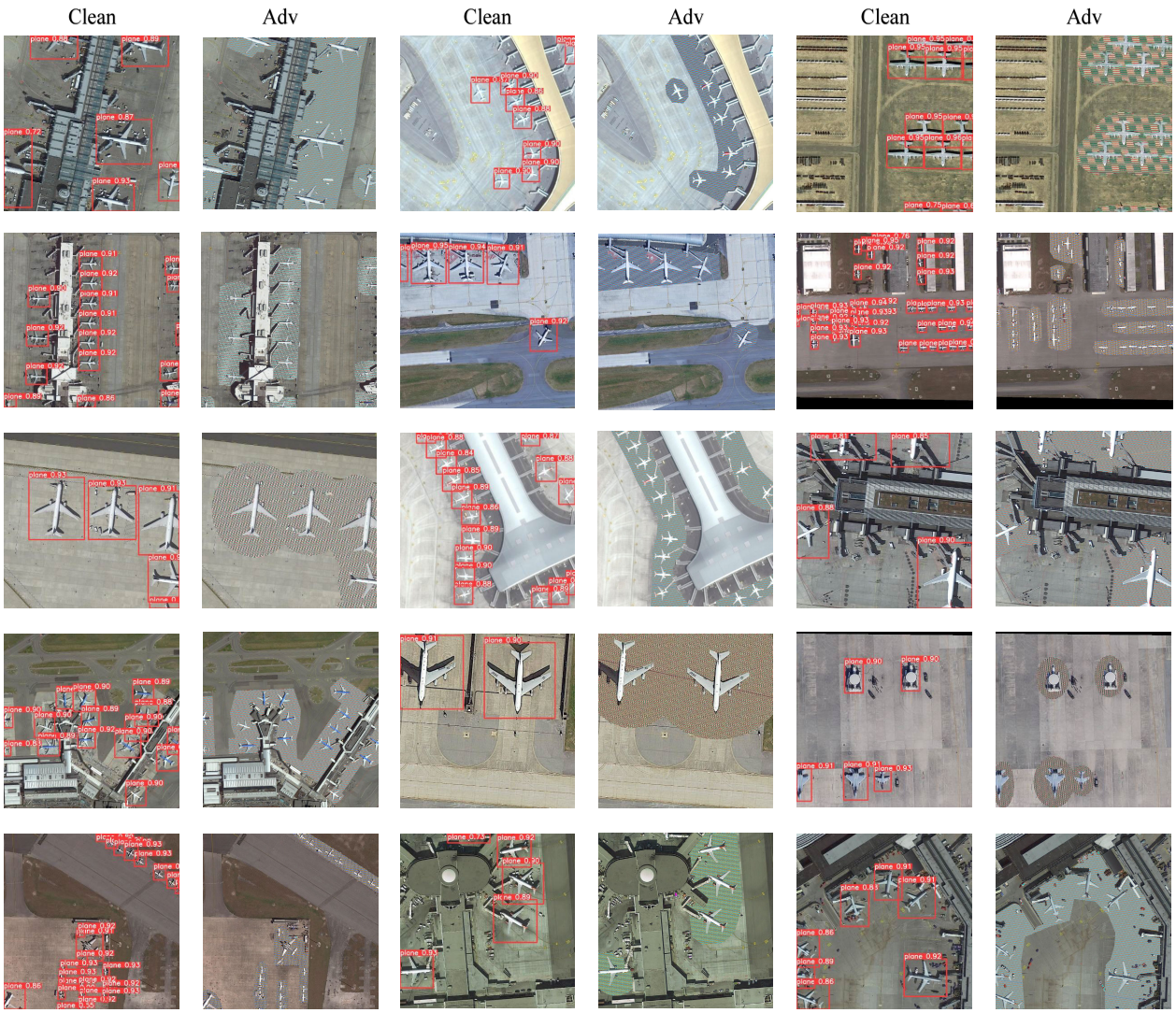


Fig. 3. Visualizing adversarial samples in the digital domain. For the YOLOv5s model, columns 1, 3, and 5 show detection results for clean samples, while columns 2, 4, and 6 display results for adversarial samples.

effective in some cases, the transferability of adversarial samples is weak. Transformer-based object detectors are notably resilient to attacks, likely due to their global feature extraction and self-attention mechanisms [74], making them difficult to compromise. Similarly, YOLOv5l demonstrates higher resistance to adversarial examples compared to YOLOv5s and YOLOv5n, attributed to its deeper network structure and richer feature learning capabilities [4]. In contrast, Cascade-R-CNN's defense against adversarial examples is limited, possibly due to its cascading structure that magnifies interference across stages [73].

Fig. 3 visualizes our experimental results, demonstrating that the adversarial textures generated by our method maintain high consistency with the background style. We modified the shape of the textures based on the image content and manually designed masks to apply them to the backgrounds, validating their extensibility. The results show that changes in shape, position, and size do not affect the attack effectiveness of the adversarial textures.

2) Attack in Physical World: This article introduces a framework primarily designed for physical attacks, and extensive experiments validate its effectiveness. We conducted experiments with five aircraft models during the validation process, and their parameters, including length and width (unit: centimeters), are presented in Fig. 4.

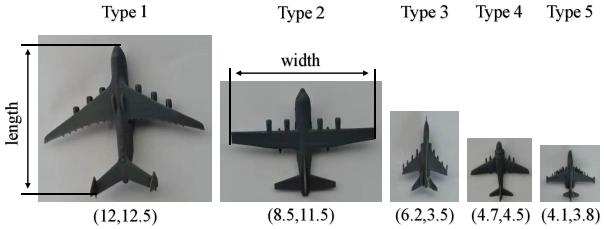


Fig. 4. Aircraft model employed in the physical attack experiments. Five types are represented, with numerical values below each image denoting the aircraft model's length and width. For example, (12, 12.5) signifies a length of 12 cm and a width of 12.5 cm for the aircraft.

In optimal daytime conditions (clear skies, minimal shadows, and a standard shooting perspective), we employed four distinct environments and their corresponding adversarial textures from Fig. 5 to evaluate the effectiveness of physical

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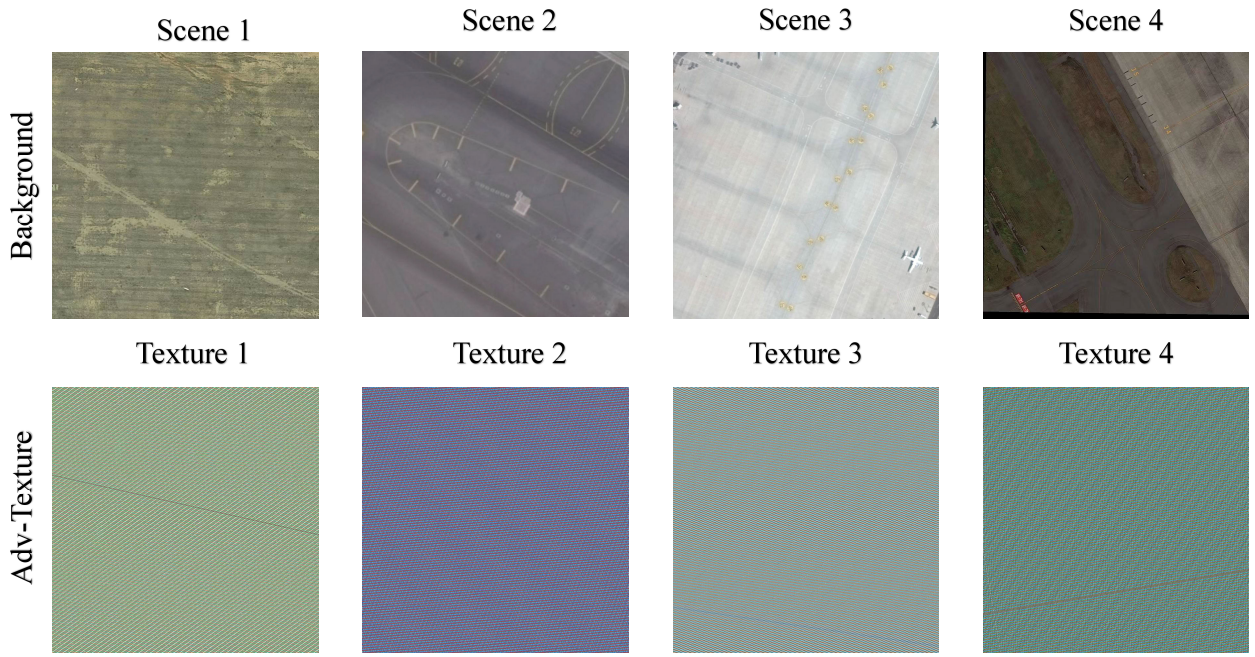


Fig. 5. We further choose four airport environments (Scene1–4, all of which are present in the Dota dataset) and their corresponding adversarial textures (Texture 1–4) for an experimental assessment of the adversarial robustness of these textures in real-world conditions.

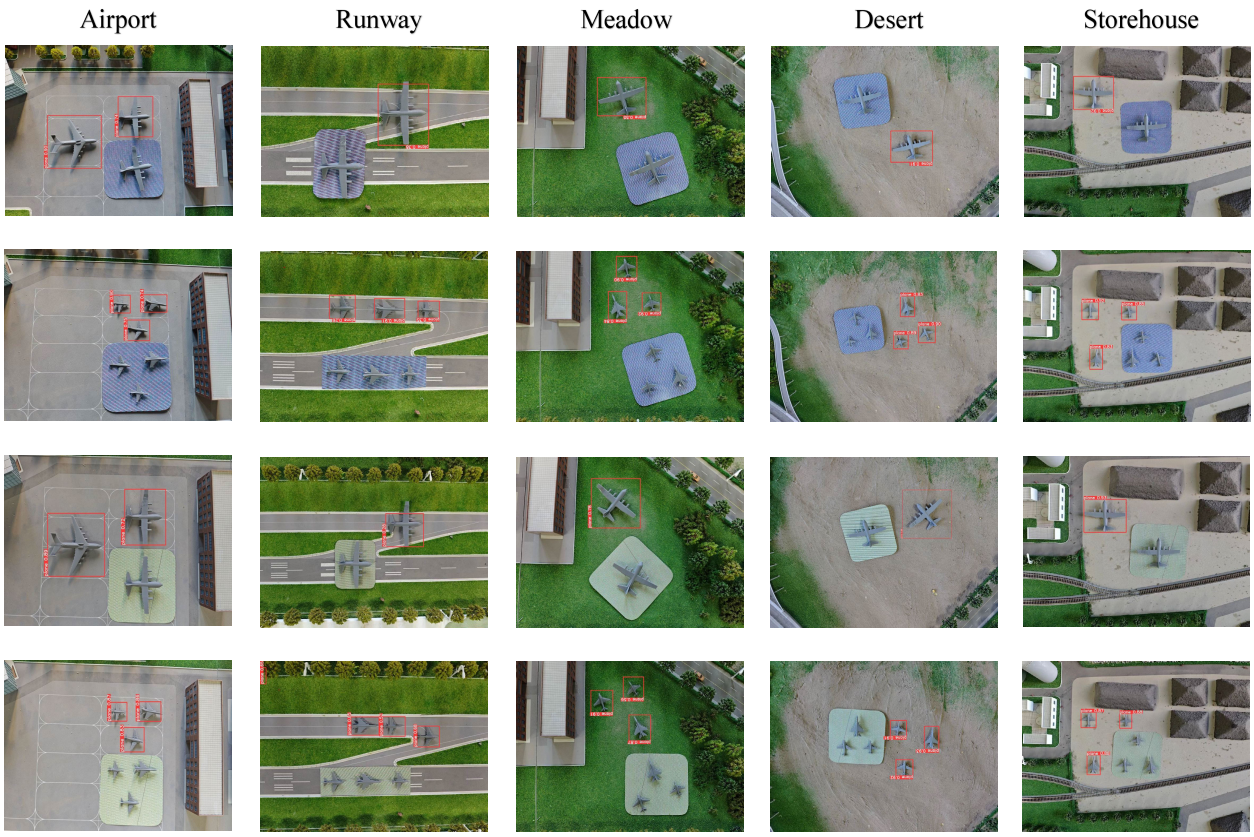


Fig. 6. We choose three distinct adversarial textures to assess the efficacy of adversarial attacks on the YOLOv5s model across five different scenarios.

attacks. We recorded the average confidence levels for ten aircraft (1 of Type 1, 2 of Type 2, 3 of Type 3, 2 of Type 4, and 2 of Type 5) using a confidence threshold of 0.2, indicating that detections below this threshold would be disregarded. The experimental outcomes are summarized in Table II, adversarial textures significantly lower the detector's confidence in aircraft detection results. Fig. 6 illustrates that our adversarial textures successfully targeted aircraft from five distinct models of

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TABLE III

INFLUENCE OF THE ADVERSARIAL TEXTURE RADIUS $r_{p'}$ AND THE TRANSPARENCY (OVERLAY RATIO) r_o OF ADVERSARIAL TEXTURE ON THE EFFICACY OF ADVERSARIAL SAMPLE ATTACKS

		Yolov5s	Yolov5n	Yolov5l	Faster R-CNN	Cascade R-CNN	Mask R-CNN	Swin-Transformer
$r_{p'}$	2	0.54%	0.24%	5.66%	1.04%	1.24%	0.70%	4.74%
	1.5	1.71%	2.46%	8.13%	1.35%	0.93%	1.76%	2.37%
	1	46.39%	48.86%	52.49%	39.86%	6.40%	18.32%	17.44%
	0.8	69.89%	73.70%	77.65%	74.74%	36.74%	78.04%	78.74%
r_o	1	1.52%	0.22%	27.83%	2.80%	3.20%	3.61%	6.50%
	0.8	7.73%	0.22%	53.74%	6.83%	12.28%	15.58%	30.96%
	0.6	38.05%	17.57%	76.51%	12.94%	37.87%	50.15%	82.35%
	0.4	63.38%	64.45%	89.89%	57.25%	63.78%	73.28%	92.26%

TABLE IV

AT A BANK ANGLE OF APPROXIMATELY 30°, EACH AIRCRAFT EMPLOYS YOLOV5S TO DETECT CONFIDENCE LEVELS ACROSS DIVERSE SCENARIOS

		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Average
Scene 1	Clean	0.91	0.88	0.87	0.87	0.91	0.89	0.91	0.87	0.91	0.92	**0.89**
	Texture 1	0.47	0.40	0.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.12**
	Texture 2	0.00	0.06	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 3	0.00	0.68	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.01	**0.07**
	Texture 4	0.00	0.61	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.09**
Scene 2	Clean	0.92	0.91	0.91	0.89	0.57	0.86	0.88	0.38	0.86	0.81	**0.80**
	Texture 1	0.21	0.25	0.14	0.01	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 2	0.01	0.57	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.11**
	Texture 3	0.06	0.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.08**
	Texture 4	0.00	0.00	0.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.08**
Scene 3	Clean	0.90	0.91	0.83	0.81	0.87	0.85	0.87	0.77	0.53	0.42	**0.78**
	Texture 1	0.00	0.73	0.00	0.43	0.00	0.00	0.00	0.00	0.00	0.01	**0.12**
	Texture 2	0.00	0.28	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	**0.03**
	Texture 3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	Texture 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
Scene 4	Clean	0.92	0.91	0.89	0.90	0.90	0.91	0.76	0.87	0.86	0.90	**0.88**
	Texture 1	0.30	0.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.07**
	Texture 2	0.01	0.55	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.05**
	Texture 3	0.00	0.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.07**
	Texture 4	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.07**

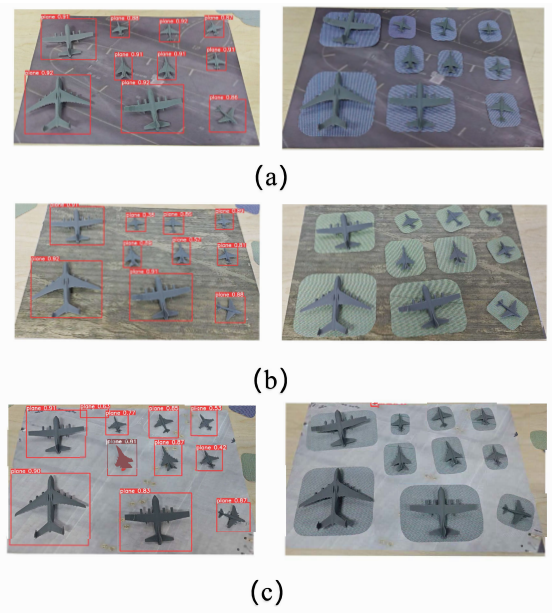


Fig. 7. Adversarial Attack Effectiveness on YOLOv5s from an oblique perspective. (a)–(c) Demonstrate our attack effects in different scenarios.

aircraft across five real-world environments (airport, runway, meadow, desert, and storehouse). We maintained the structural integrity of the surroundings, ensuring consistency in texture style and background tone, achieving a notable level of stealthiness in the attacks.

3) Attack Robustness: To further confirm the robustness of our generated adversarial textures, we randomly selected an optimized texture and assessed its attack effectiveness across the entire dataset in the data domain. Different radii and overlay ratios were varied to simulate practical texture application methods such as printing or spraying, considering texture range size and fading. Table III shows a positive correlation between attack effectiveness and texture coverage range. The optimal balance was achieved when the length of the adversarial texture axis was 1.5 times the aircraft's length-to-width ratio, as depicted in Fig. 3. Therefore, we investigated the impact of the adversarial texture superposition ratio on the attack effectiveness when confronting the length of the texture axis with an aspect ratio of 1.5. Even as the adversarial texture fades, it retains a certain attack effectiveness level. Therefore, the adversarial textures generated by our approach exhibit temporal persistence.

In the physical domain, we tested robustness in real-world environments, including skewed angles, various filters, diverse lighting, and scenarios with incomplete aircraft coverage. Table IV displays results from testing at an approximately

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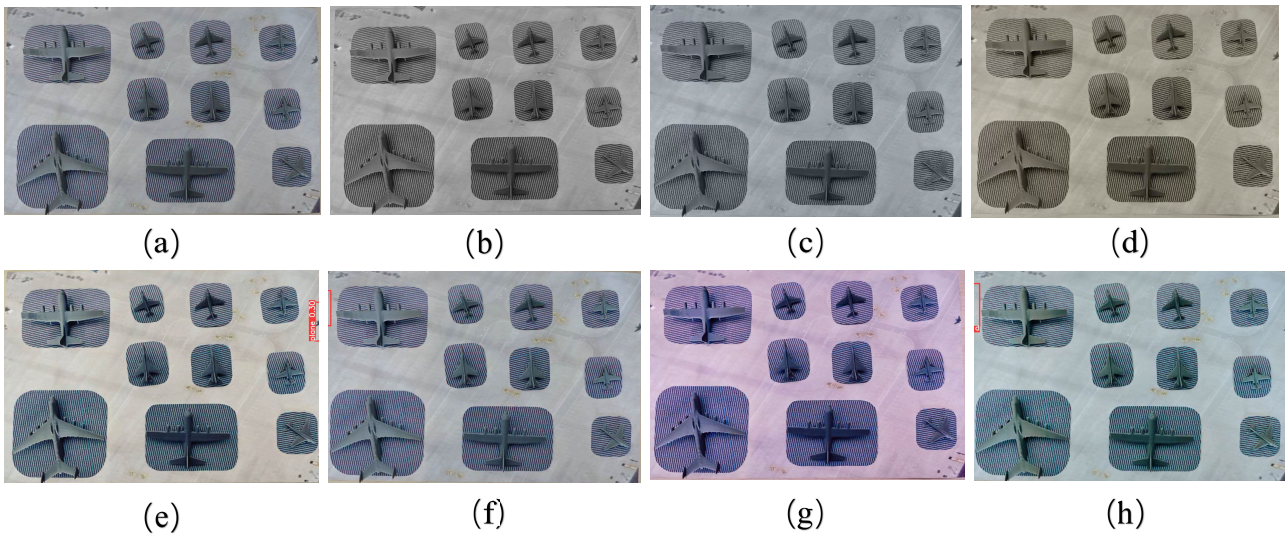


Fig. 8. To further validate the robustness of adversarial textures, we assessed their attack effectiveness under eight different shooting conditions. Across various shooting modes, adversarial textures consistently rendered the detector ineffective. (a)–(h) Showcase the attack effects under eight different filters.

30° angle relative to the horizontal line, with recorded aircraft confidence levels. The adversarial texture proved insensitive to skewed angles and retained attack effectiveness at higher inclinations shown in Fig. 7. This may be because our method generates adversarial textures that are decoupled from their shape. Thus, even after deformation at larger tilt angles, the attack effectiveness remains intact. The effectiveness likely depends on the finer details of the texture, so preserving detailed information ensures optimal attack performance. Fig. 8 verifies attack effectiveness under different filters using a randomly chosen background and adversarial texture, highlighting consistent attack potency under diverse shooting conditions. Fig. 9(a)–(c) demonstrates validation under different shapes and scenarios of incomplete coverage, while Fig. 9(d)–(f) depicts verification under various lighting conditions. The experimental results suggest that the effectiveness of adversarial textures is not solely reliant on texture shape, color, or brightness. Instead, it may be linked to more intricate internal texture structures. Additionally, the adversarial attack performance remains effective even under conditions of lower image quality. As depicted in the experimental display figures, we did not standardize the image resolution during testing; nonetheless, the adversarial textures maintained their attack effectiveness across images of varying resolutions.

In summary, our adversarial textures demonstrate robustness in both digital and physical domains, showcasing potent attack effectiveness across diverse environments and conditions. In practical implementations, users have the flexibility to select appropriate shapes and sizes, illustrating expandability. Furthermore, these textures demonstrate enduring durability.

4) Impact of Feature Alignment Layer: In this section, we systematically examine the influence of different feature alignment layers on both the attack performance and the visual effects of adversarial textures. The VGG19 network primarily comprises five convolutional blocks and three fully connected layers. The first two convolutional blocks capture low-level features, the middle block extracts mid-level features, and the final two blocks capture high-level features with fine-grained semantic information [67]. In generating adversarial textures, we aim to match the background environment's color tone for visual harmony rather than replicating its complex features. Therefore, our experiments are divided into two parts: aligning

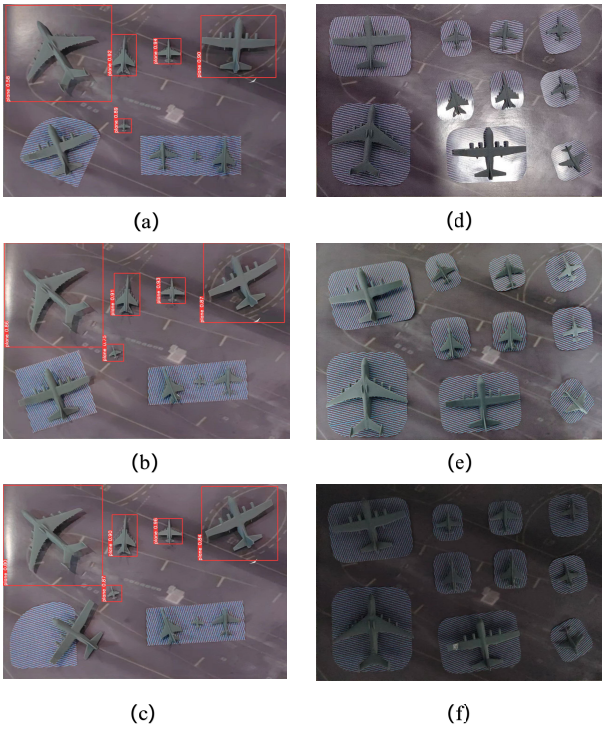


Fig. 9. We investigated how the shape and coverage of adversarial textures influence their attack effectiveness on YOLOv5s. As shown in (a)–(c), the shape and orientation of adversarial textures do not impact their attack effectiveness. Moreover, even when the aircraft is not fully covered, the adversarial textures still demonstrate effectiveness. In (d)–(f), we demonstrate the attack effectiveness of adversarial textures under altered lighting conditions. The figures highlight the robustness of adversarial textures across different lighting conditions.

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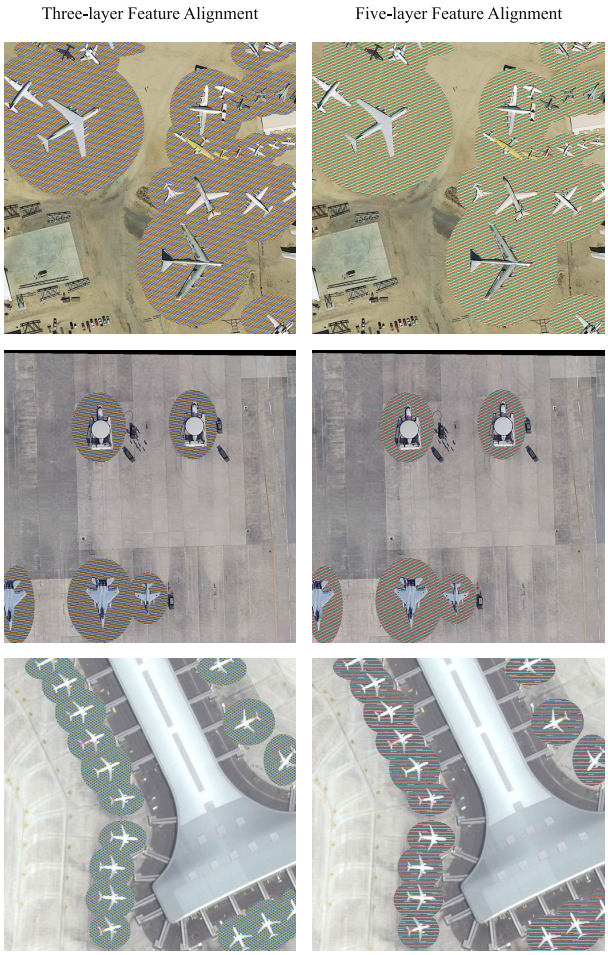


Fig. 10. Different numbers of feature alignment layers for adversarial samples, targeting YOLOv5s.

adversarial texture features using only the first three layers of VGG19 and aligning using all five layers. As depicted in Table I, including three or five layers of feature alignment in constructing the adversarial texture generator showed no significant impact on its attack performance. Nevertheless, as illustrated in Fig. 10, it is evident that adversarial textures generated with five layers of feature alignment display superior visual effects, characterized by a more intricate texture structure and well-aligned color tones. Since high-level features contain global information, aligning them may shift the color tone of the adversarial textures. The analysis indicates that augmenting the number of feature alignment layers can enhance the texture generator's expressive capacity, allowing it to more effectively capture subtle features and style information from the original images. Conversely, reducing the number of feature alignment layers can mitigate computational burdens.

V. Conclusion

In this article, we propose an expandable adversarial texture generation method for aviation image target detection in airport environments. Our primary aim is to enhance the visual coherence and scalability of the adversarial texture while avoiding interference with the aircraft surface, facilitating large-scale application in real-world scenarios. Our approach leverages the translation invariance property of a convolutional network and adjusts the data distribution of the adversarial texture feature map by modifying the airport feature map's data distribution during texture generation. This achieves shape-independent texture adversarialness, enabling arbitrary scalability and aligning the texture with the airport environment. To transition the adversarial texture seamlessly from the digital to the physical domain, we comprehensively model the real environment in the digital space, incorporating techniques like affine transformations, random cropping, and Gaussian blurring. Experimental results demonstrate excellent performance in attack success rate, visual coherence, and robustness, with promising outcomes in adversarial transferability.

For future work, we plan to develop more robust adversarial textures adaptable to diverse scenarios, extreme environments, and high dynamic scenes, aiming to advance the resilience and security of machine learning models.

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